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6. Modelling cold regions



Learning Objectives

After today's lecture you should be able to:

- Understand the significance and occurrence of frozen soils globally
- Understand the how snowmelt dominates the hydrological cycle in seasonally frozen soils
- Understand concepts of specific and latent heat and calculate the equilibrium state of ice and water combined in an isolated system.
- Understand the thermal properties (heat capacity and thermal conductivity) of soils
- Describe how water, ice and air are partitioned with the soil pore space
- Understand the soil freezing characteristic curve, and it's similarity to the SMC
- Appreciate the challenges of physically based modelling of frozen soils

Modelling

What is a model?

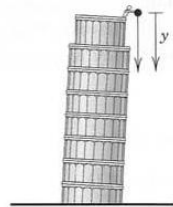
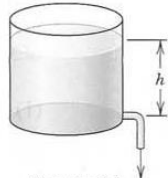
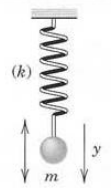
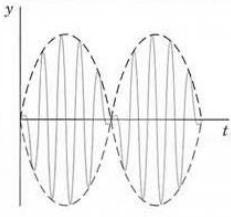
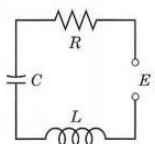
A model may be defined as a selected simplified version of a real system and phenomena that takes place within it, which approximately simulates the system's excitation-response relationships that are of interest. ” **Bear & Cheng, 2010**

Physical model



<http://www.nrc-cnrc.gc.ca/eng/projects/chc/temiscaming-dam.html>

Mathematical model

 Falling stone $y'' = g = \text{const.}$ (Sec. 1.1)	 Parachutist $mv' = mg - bv^2$ (Sec. 1.2)	 Water level h Outflowing water $h' = -k\sqrt{h}$ (Sec. 1.3)
 Displacement y Vibrating mass on a spring $my'' + ky = 0$ (Secs. 2.4, 2.8)	 Beats of a vibrating system $y'' + \omega_0^2 y = \cos \omega t, \quad \omega_0 = \omega$ (Sec. 2.8)	 Current I in an RLC circuit $LI'' + RI' + \frac{1}{C}I = E'$ (Sec. 2.9)

Kreysig, E. 2011, Advanced Engineering Math. p. 3

A perfect model?

If you want to create a perfect model of the earth...

... you would have to create a perfect duplicate of the earth



There are no perfect models! A model is always simplified.

There are no universal models! A model should be designed to address a very specific problem.

Why do we need environmental models?

- We can't measure everything we would like to know
- What we can measure, we measure imperfectly
- We therefore use models to extract useful information from imperfect and incomplete data. We are looking to extend our knowledge in:



Time: especially into the future, but also into the past, or between observations

Space: interpolation between measurements, extrapolation beyond measurements and

Scale: translating information from one scale to another , over a wide range of scales in space (e.g. soil pore scale up to continental scale) and time (e.g. days to years)

How are models useful?

There are two ways the models can be useful:

1. Predictive modelling

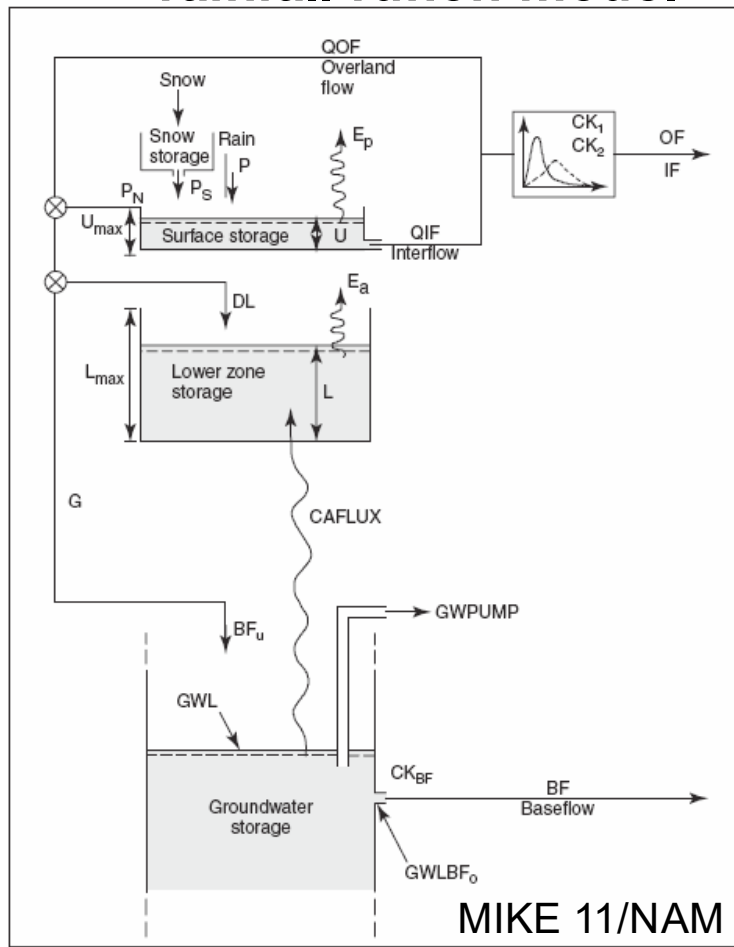
- This is most obvious purpose – if a model works well, we can use it to predict the future
- If we could do this, the benefits are obvious: prepare for natural disasters ahead of time, determine the best course of action, make wise investments
- Unfortunately, we can only do this well to a limited extent when we deal with environmental systems

2. Explanatory modelling

- Less obvious, but often more useful and practical.
- This is the process of using models to test and further our understanding of how a system works.
- Knowing how a system works doesn't always mean you can make accurate quantitative predictions, but it does let you know qualitatively how the system might respond to a particular change or event.

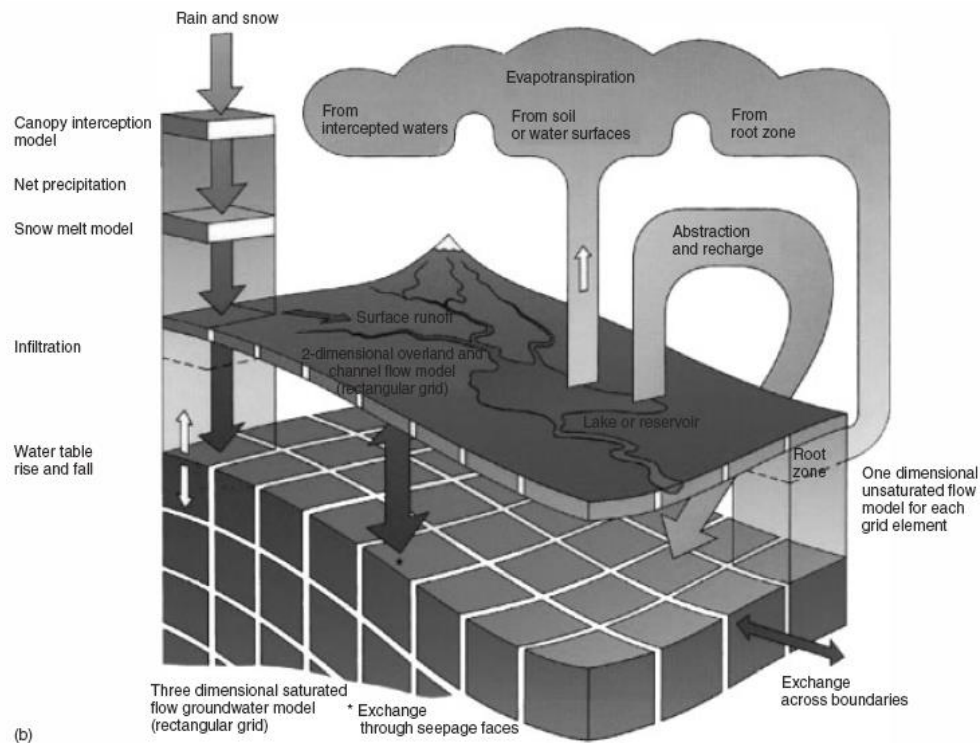
Types of hydrological model

A lumped, conceptual rainfall-runoff model



More empirically based

A distributed, integrated modelling system



MIKE SHE

More physically based

CONCEPTUAL MODELS

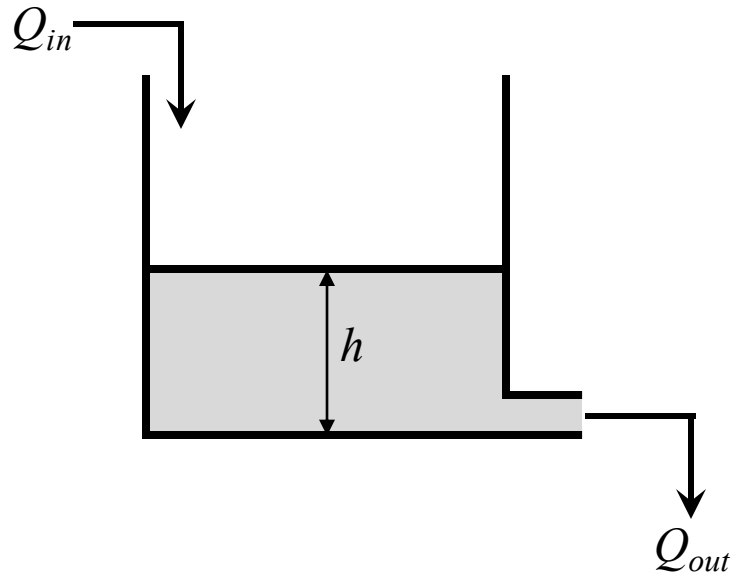
Conceptual models

- Recall that models may be classified on a spectrum which extends from **Empirical** (Inductive, data-based) at one end to **Physically Based** (deductive, mechanistic) at the other
- The term “conceptual model” can be used to refer to a model which is close to the middle of this spectrum.

A conceptual model:

- Has a model structure with some physical meaning, with process representations that can be interpreted
- Has parameters that can be calibrated against observed data
- Seeks to balance *physical realism* with *parsimony*
- Typically, conceptual models can be shown diagrammatically as one or more simple stores, or “buckets”
- One physical principle which applies rigidly to all conceptual models is that of continuity ~ i.e. mass is not created or destroyed.

The water balance of a leaky bucket



Variables:

Q_{in}	Inflow [$\text{m}^3 \text{d}^{-1}$]
Q_{out}	Outflow [$\text{m}^3 \text{d}^{-1}$]
h	Water level [m]
A	Base area [m^2]
ρ	Density of water [kg m^{-3}]
t	Time [d]

Continuity: Change in Mass = Mass In – Mass Out

$$\text{Change in Mass} = \Delta V \cdot \rho = \Delta h \cdot A \cdot \rho$$

$$\text{Mass In} = Q_{in} \cdot \Delta t \cdot \rho$$

$$\text{Mass Out} = Q_{out} \cdot \Delta t \cdot \rho$$

Therefore:

$$\frac{\Delta h}{\Delta t} = \frac{Q_{in} - Q_{out}}{A}$$

As $\Delta t \rightarrow 0$, then

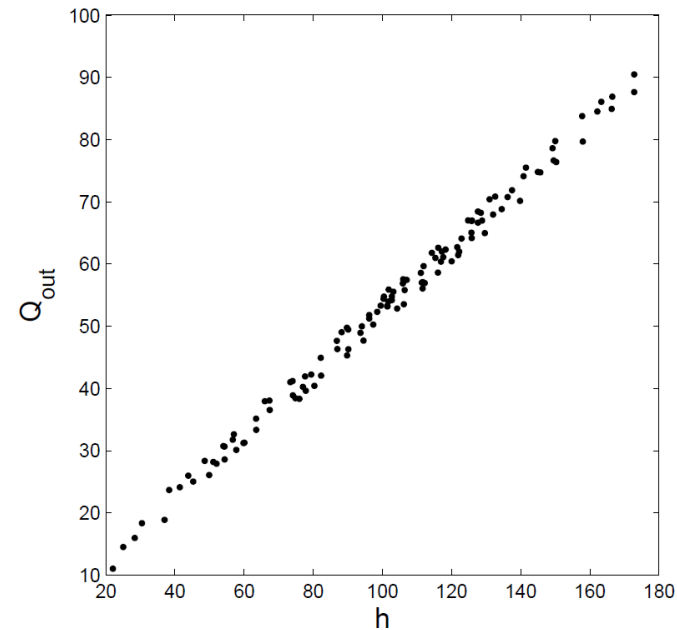
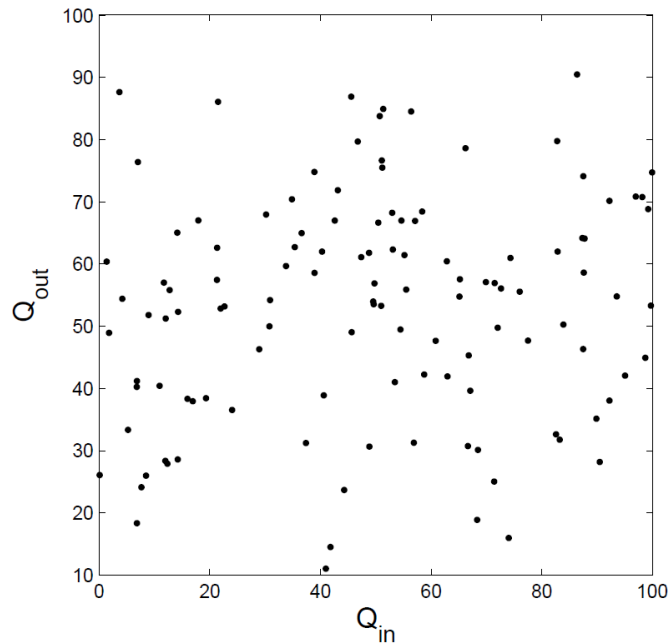
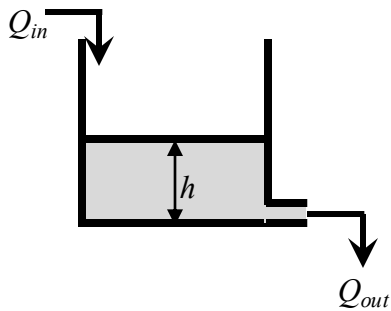
$$\frac{dh}{dt} = \frac{Q_{in} - Q_{out}}{A}$$

Now check
dimensions

Discharge from the bucket

If we are representing a natural (as opposed to managed) system, we can sometimes express the discharge from the bucket as a function of some other property.

What do you think is an sensible property of this system to related the discharge to? Consider the (hypothetical) observations below:



Discharge from the bucket

In this example, it would make sense to set

$$Q_{out} = k.h$$

Which describes the line shown in the graph:

Hence we can rewrite the governing equation as

$$\frac{dh}{dt} = \frac{Q_{in} - k.h}{A}$$

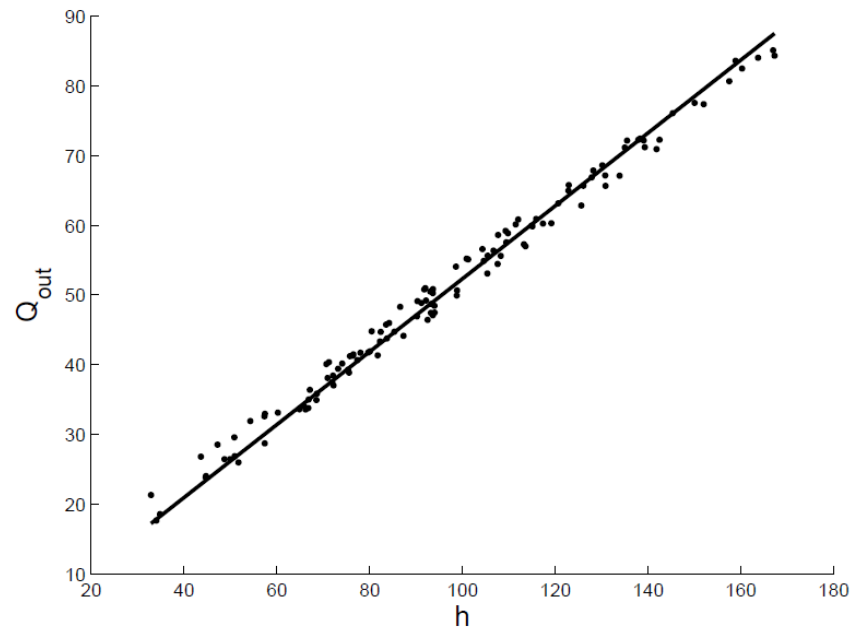
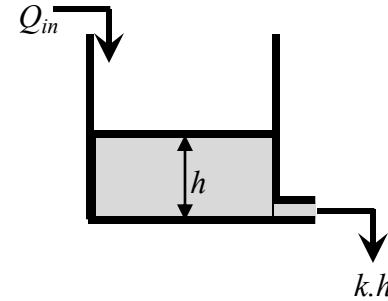
So, here we have the **storage** relationship:

$$V = h.A \quad [\text{m}^3]$$

or $\Delta V = (\Delta h).A \quad [\text{m}^3]$

And the **transmission** relationship:

$$Q_{out} = k.h \quad [\text{m}^3 \text{ d}^{-1}]$$



Ordinary Differential Equations

Consider the governing equation that we have just derived

$$\frac{dh}{dt} = \frac{Q_{in} - k.h}{b + 2ha} \quad (1)$$

This is a first order equation, because it's highest derivative is a first derivative.

For example $\frac{d^2y}{dx^2} = ax + by$ is a second order equation.

It is an ordinary differential equation, because there is only one independent variable. What is the independent variable?

For example $\frac{\partial y}{\partial t} = D \cdot \frac{\partial^2 y}{\partial x^2}$ is not an ordinary differential equation. What is it?

A linear ODE is one for which there are no products (i.e. non-first order powers) or non-linear functions of the dependent variable or its derivatives.

What is the dependent variable in (1)? Is this a linear ODE?

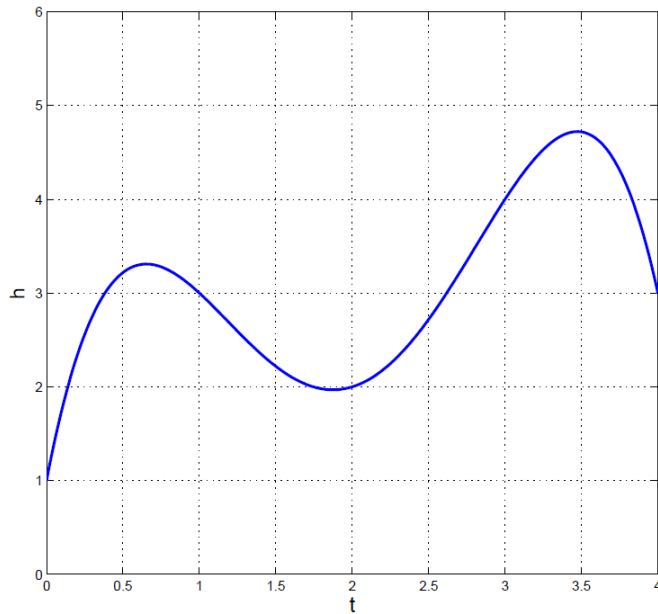
An example of a linear ODE is $\frac{dh}{dt} = \frac{Q_{in} - k.h}{A}$ and a linear PDE is $\frac{\partial y}{\partial t} = D \cdot \frac{\partial^2 y}{\partial x^2}$

Numerical solutions to ODEs

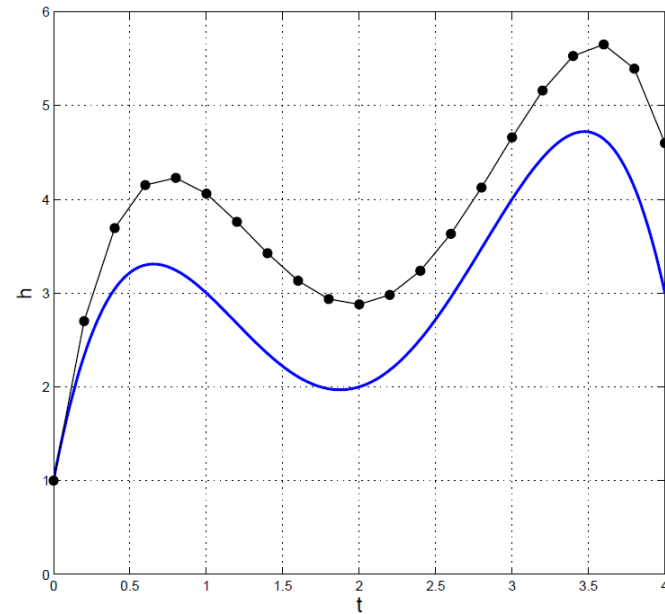
Our objective is, given some initial value of h , to find how h varies as a function of time.

We will apply numerical methods to attempt to approximate the true answer.

TRUE ANSWER

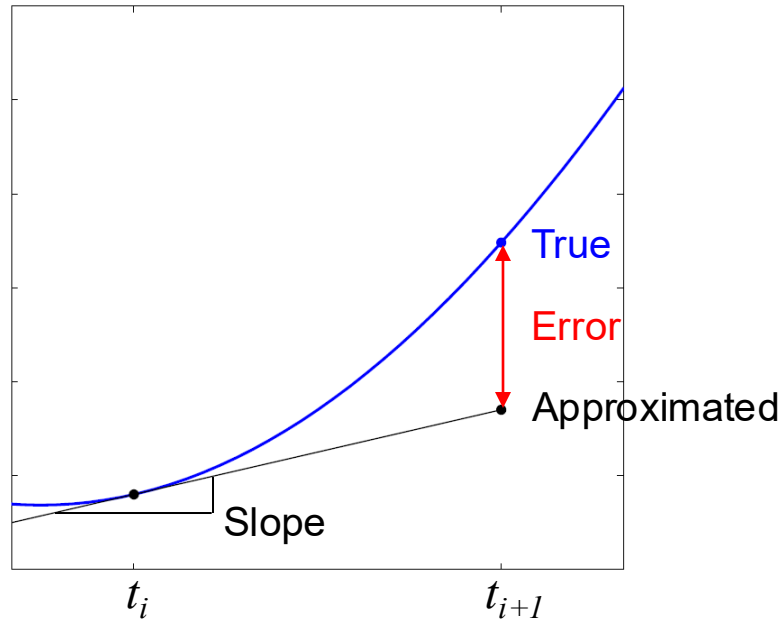


NUMERICAL APPROXIMATION



Euler's method

Euler's method is the simplest to apply to this problem



The method can be described:

$$\text{New value} = \text{Old value} + \text{slope} \times \text{step size}$$

In mathematical notation: $h_{i+1} = h_i + f(t_i, h_i) \cdot \Delta t$

where $f(t_i, h_i) = \left. \frac{dh}{dt} \right|_i$

For the problems we are interested in, our governing equations define $f(t_i, h_i)$, for example see Equation (1) on slide 23.

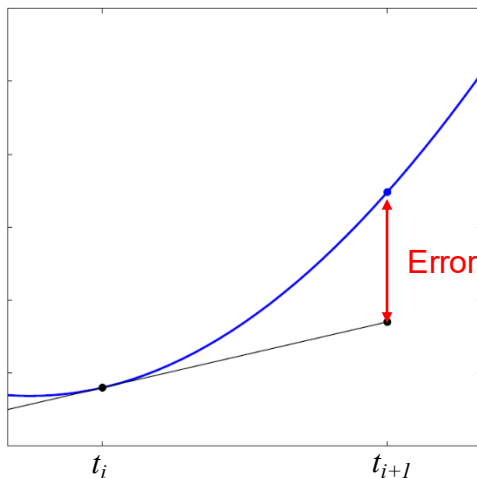
Euler's method

Euler's method is simple, but can result in large truncation errors.

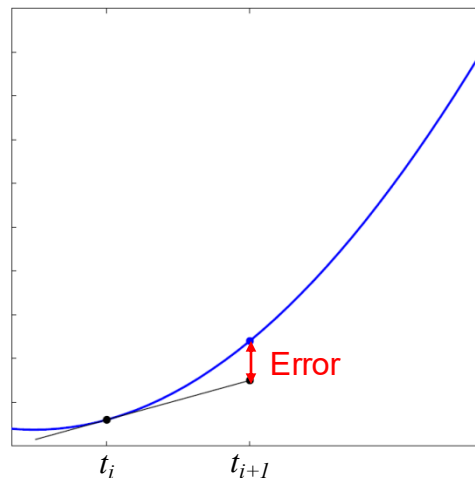
Truncation errors can generally be minimized by reducing the size of the step.

Truncation errors in this method also cumulate (or propagate) as the solution marches forward in time.

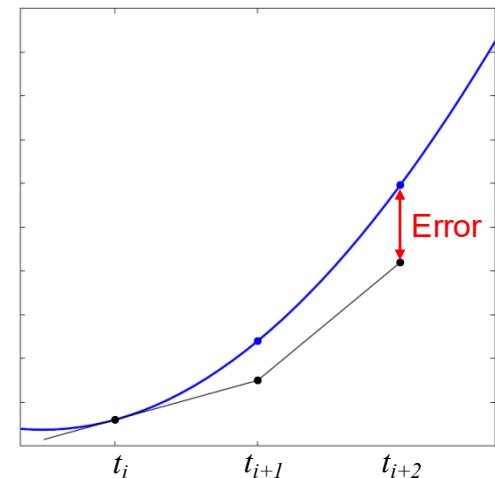
Original approximation



Half the step size



Propagate to 2nd step



Other methods for ODEs

There are various other methods for integrating ODEs, which are nicely described by *Chapra and Canale, 1998. Numerical Methods for Engineers. McGraw Hill*. At this stage, we have two considerations:

1. Accuracy – i.e. minimizing the truncation error

2. Efficiency – i.e. minimizing runtime.

Many practical problems can be, and are, addressed very effectively using Euler's method.

HBV-SASK Model

- HBV-SASK is a conceptual hydrological model that was put together by **Dr Saman Razavi** (these slides and the model are credited to Dr Razavi)
- HBV-SASK is an interpretation of Hydrologiska Byråns Vattenbalansavdelning (HBV) model (*Lindström et al., 1997, JoH*), providing a simple dynamical system model for educational and exploratory purposes.
- HBV-SASK has been developed for a range of learning, teaching, and research purposes, in particular for advancing the field of parameter identification, sensitivity and uncertainty analysis.
- HBV-SASK's implementation is modular, with each module simulating a different hydrologic process, thereby allowing the user to easily investigate the different model flux and state variables.
- HBV-SASK is available free-of-charge as part of the VARS-TOOL software package (<http://vars-tool.com/>).

History of HBV

Source: <https://hepex.irstea.fr/the-hbv-model-40-years-and-counting>



**Countries where the HBV model
has been applied**

HBV – THE MOST FAMOUS HYDROLOGICAL MODEL OF ALL? AN INTERVIEW WITH ITS FATHER: STEN BERGSTRÖM

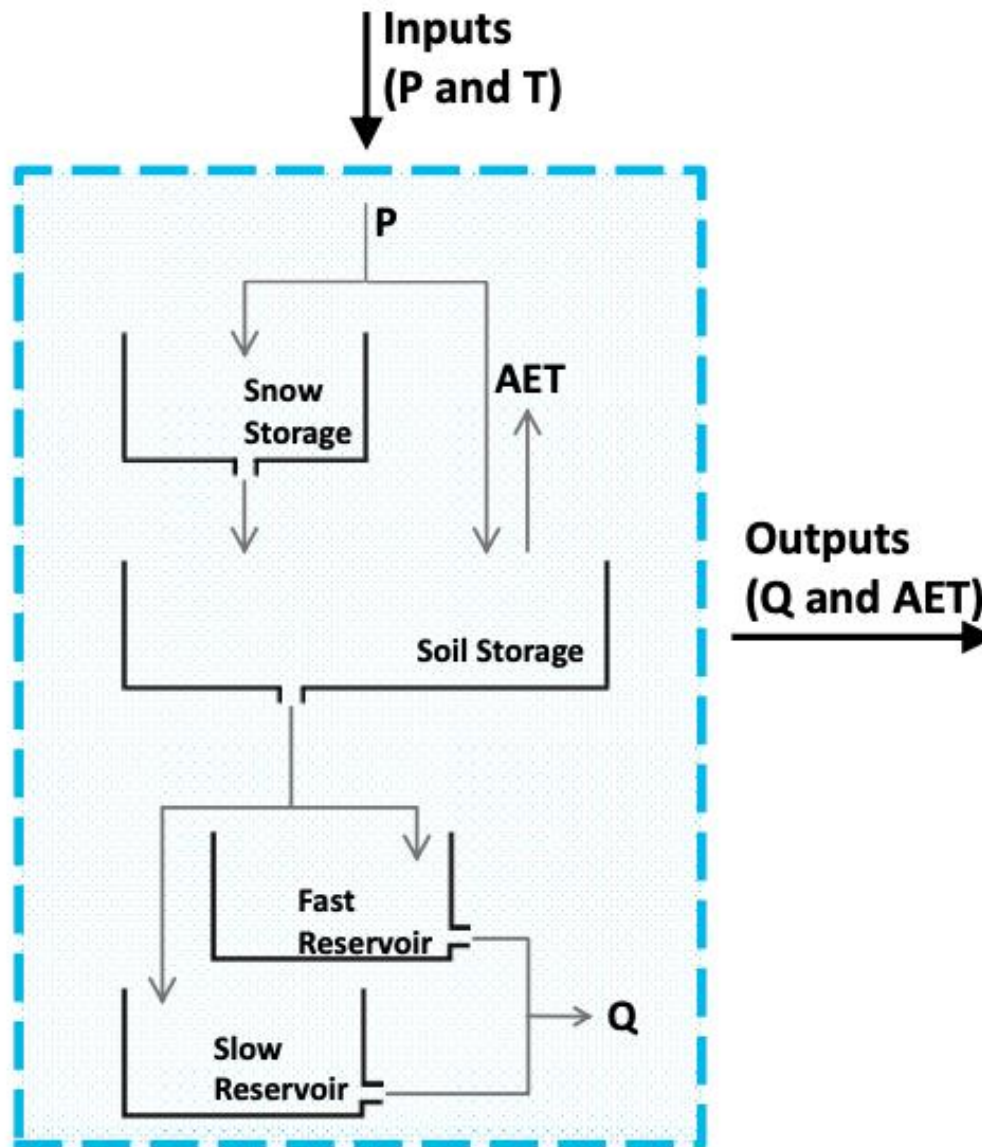
December 16, 2014 Fredrik Wetterhall 4 Comments

Contributed by Fredrik Wetterhall

Numerous are the hydrologists that has come in contact with the HBV (*Hydrologiska byråns vattenbalansavdelning*) model, either through their own hands-on experience in applying it themselves or through the vast literature on the model and its applications. The HBV model, originally developed by Sten Bergström at the Swedish meteorological and Hydrological Institute (SMHI), is one of the most used hydrological models in the world.

Next year, Sten will retire from SMHI, and to mark that occasion Sten has collected information on where the model has been applied around the world (see: [HBV_publications.pdf](#)). The reference that is most often used as the origin of the model is Bergström (1976), however the earliest versions of the model dates back to 1972. In its more than 40 years of existence, it has been used in more than 1400 publications, applied in 95 countries and used in almost 50 PhD theses.

HBV-SASK Conceptual Diagram

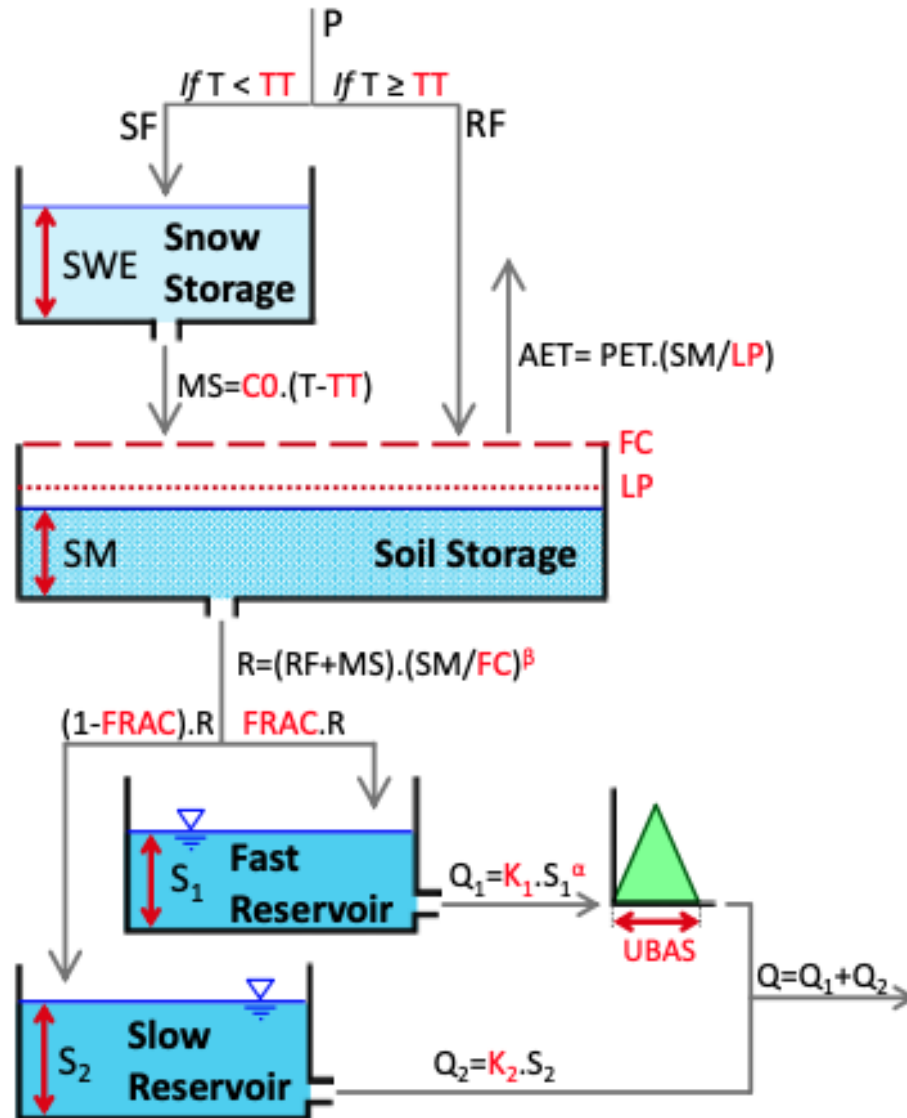


4 control volume,
each with a single
state variable
(storage)

For each control
volume we solve the
conservation
equation:

$$\frac{dS}{dt} = In - Out$$

HBV-SASK Fluxes

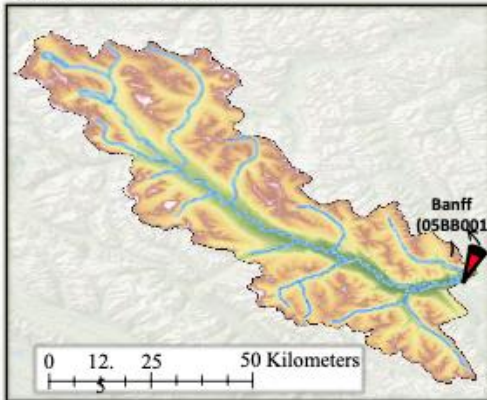


HBV-SASK Parameters

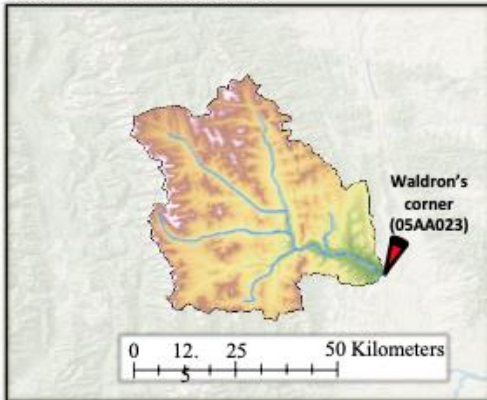
Number	Parameter Name	Lower Bound	Upper Bound	Description
1	TT	-4	4	Air temperature threshold in °C for melting/freezing and separating rain and snow
2	CO	0	10	Base melt factor, in mm/°C per day
3	ETF	0	1	Temperature anomaly correction in 1/°C of potential evapotranspiration
4	LP	0	1	Limit for PET as a multiplier to FC, i.e., soil moisture below which evaporation becomes supply limited
5	FC	50	500	Field capacity of soil, in mm. The maximum amount of water that the soil can retain
6	β (beta)	1	3	Shape parameter (exponent) for soil release equation (unitless)
7	FRAC	0.1	0.9	Fraction of soil release entering fast reservoir (unitless)
8	K1	0.05	1	Fast reservoir coefficient, which determines what proportion of the storage is released per day (unitless)
9	α (alpha)	1	3	Shape parameter (exponent) for fast reservoir equation (unitless)
10	K2	0	0.05	Slow reservoir coefficient which determines what proportion of the storage is released per day (unitless)
11	UBAS	1	3	Base of unit hydrograph for watershed routing in day; default is 1 for small watersheds
12	PM	0.5	2	Precipitation multiplier to address uncertainty in precipitation (unitless); default is 1.

HBV-SASK Case studies

Bow River Basin



Oldman River Basin



Saskatchewan River Basin



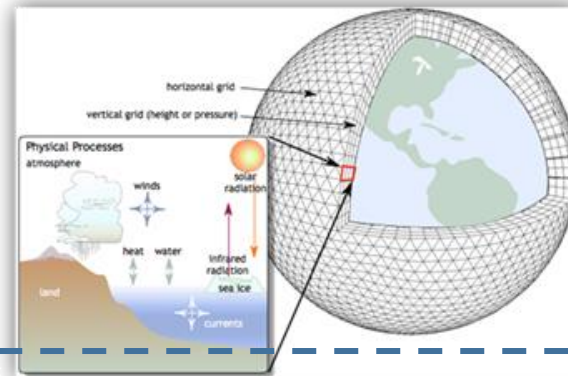
PHYSICALLY BASED MODELS

Land surface schemes

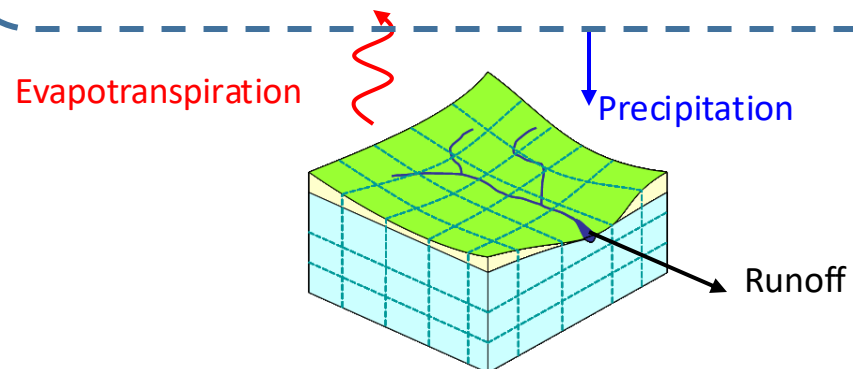
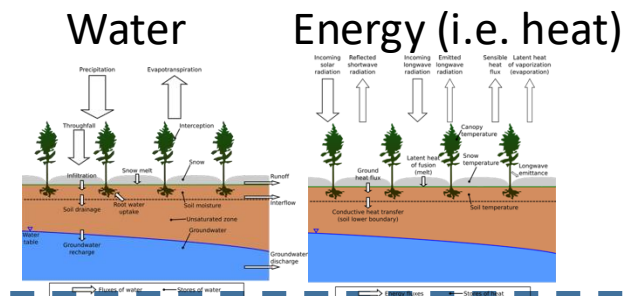
Atmospheric models (weather/climate) need to quantify the exchange of energy (and water) between the atmosphere and land

Land Surface Schemes intersect with both, simulating the vertical exchanges of water and energy.

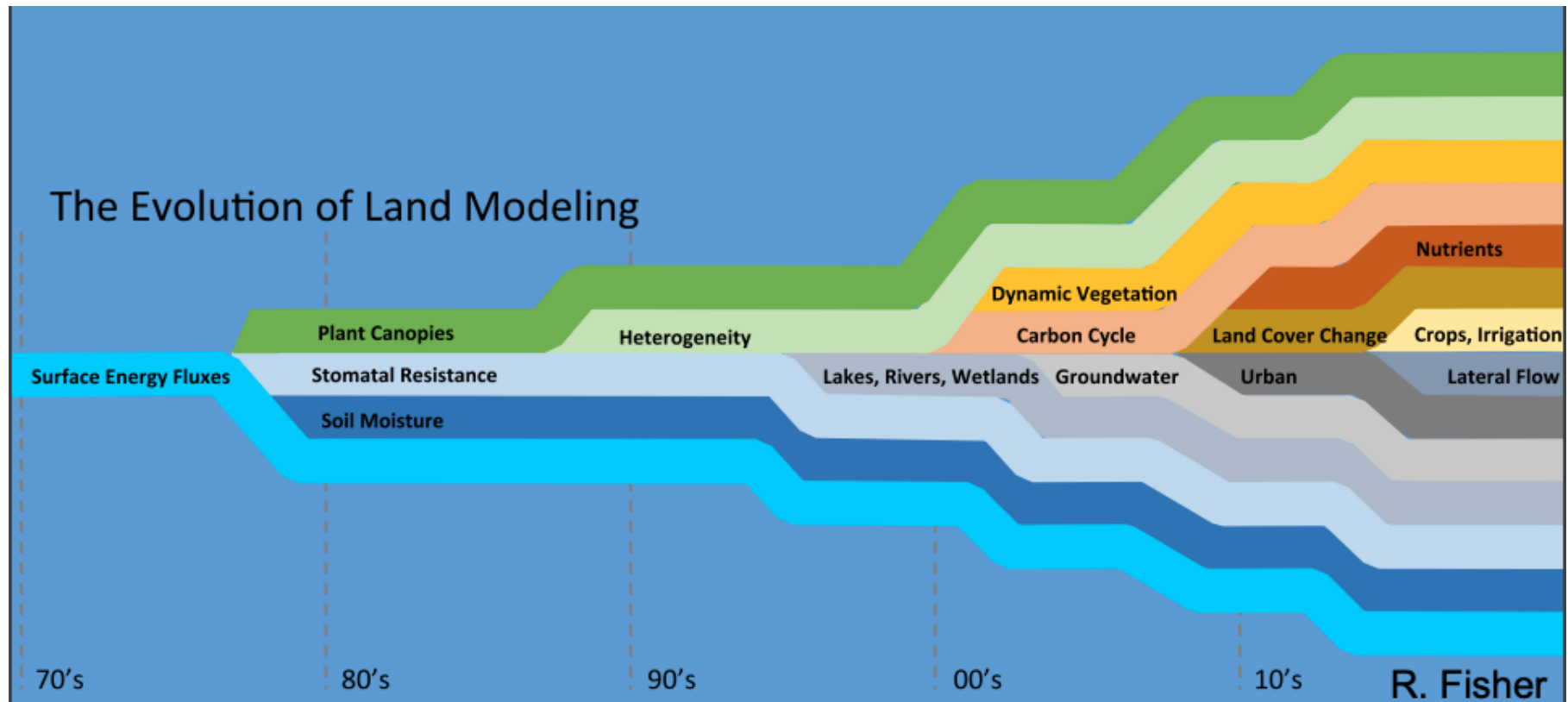
Hydrological models need to quantify the exchange of water (and energy) between the atmosphere and the land



Source: Centre for Multiscale Modelling of Atmospheric Processes)



LSS evolution in time



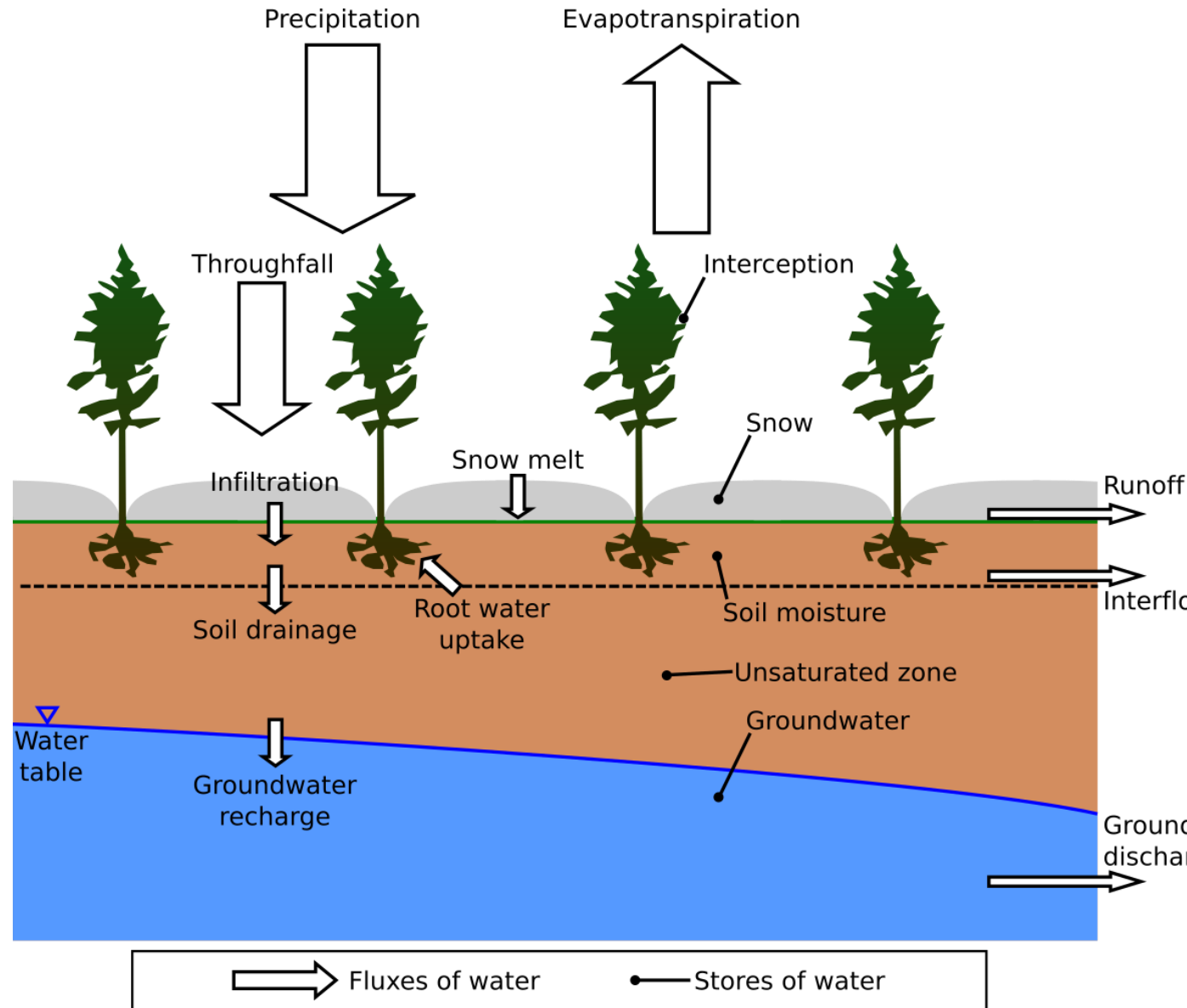
<http://www.cesm.ucar.edu/events/wg-meetings/2018/presentations/lmwg/clark.pdf>

Vertical water balance

In aggregate, the water balance equation is

$$\frac{dS}{dt} = P - E - R$$

In detail, the storage term, S (mm) and flux terms P , E and R (mm/d) all comprise further subcomponents, as shown.

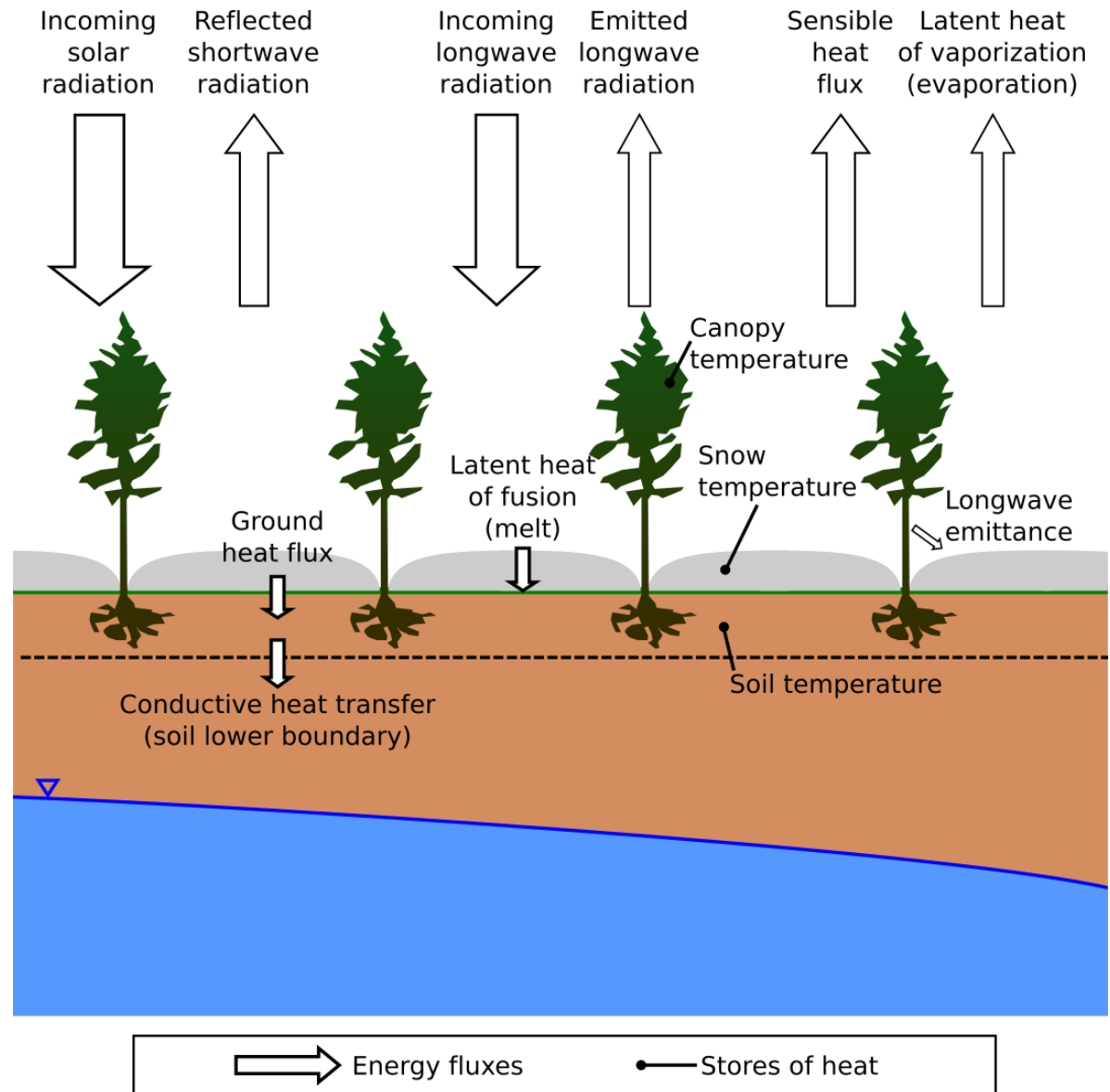


Vertical energy balance

In aggregate, the energy balance equation is

$$\frac{dU}{dt} = R_n - H - \lambda E - G$$

In detail, the storage term, U (J/m²) and flux terms R_n , H , λE and G (W/m²) all comprise further subcomponents, as shown.



Putting it all together

The coupled mass and heat transport equations look something like this:

$$\overset{\text{SMC}}{\boxed{\frac{\partial \theta_l(\psi)}{\partial t}}} + \overset{\text{SFC}}{\boxed{\frac{\rho_i}{\rho_l} \frac{\partial \theta_i(T)}{\partial t}}} = \frac{\partial}{\partial z} \left[\overset{K(\psi)}{\boxed{K(\psi)}} \left(\frac{\partial \psi}{\partial z} + 1 \right) \right] + \frac{1}{\rho_l} \frac{\partial q_v}{\partial z} - S$$

$$\boxed{C \frac{\partial T}{\partial t}} - \boxed{L_f \rho_i \frac{\partial \theta_i}{\partial t}} = \frac{\partial}{\partial z} \left[\boxed{\kappa_T \frac{\partial T}{\partial z}} \right] - \rho_l c_l \frac{\partial q_l T}{\partial z}$$

Sensible heat
Latent heat
Thermal conductivity

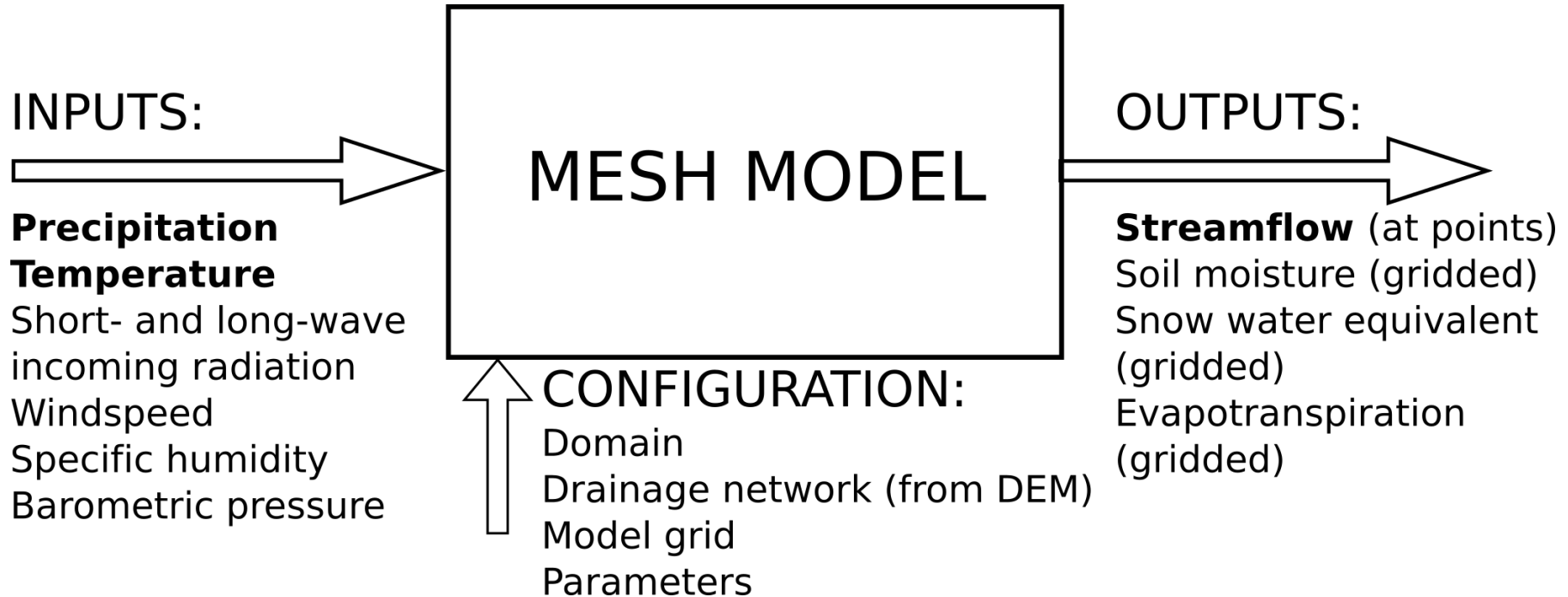
In other words, these two equations incorporate pretty much everything we have discussed today (and some stuff we haven't – vapor flow!)

These equations are solvable. The problem is **quantifying the hydraulic properties** under different temperature/moisture conditions. This is a major scientific challenge

The MESH model

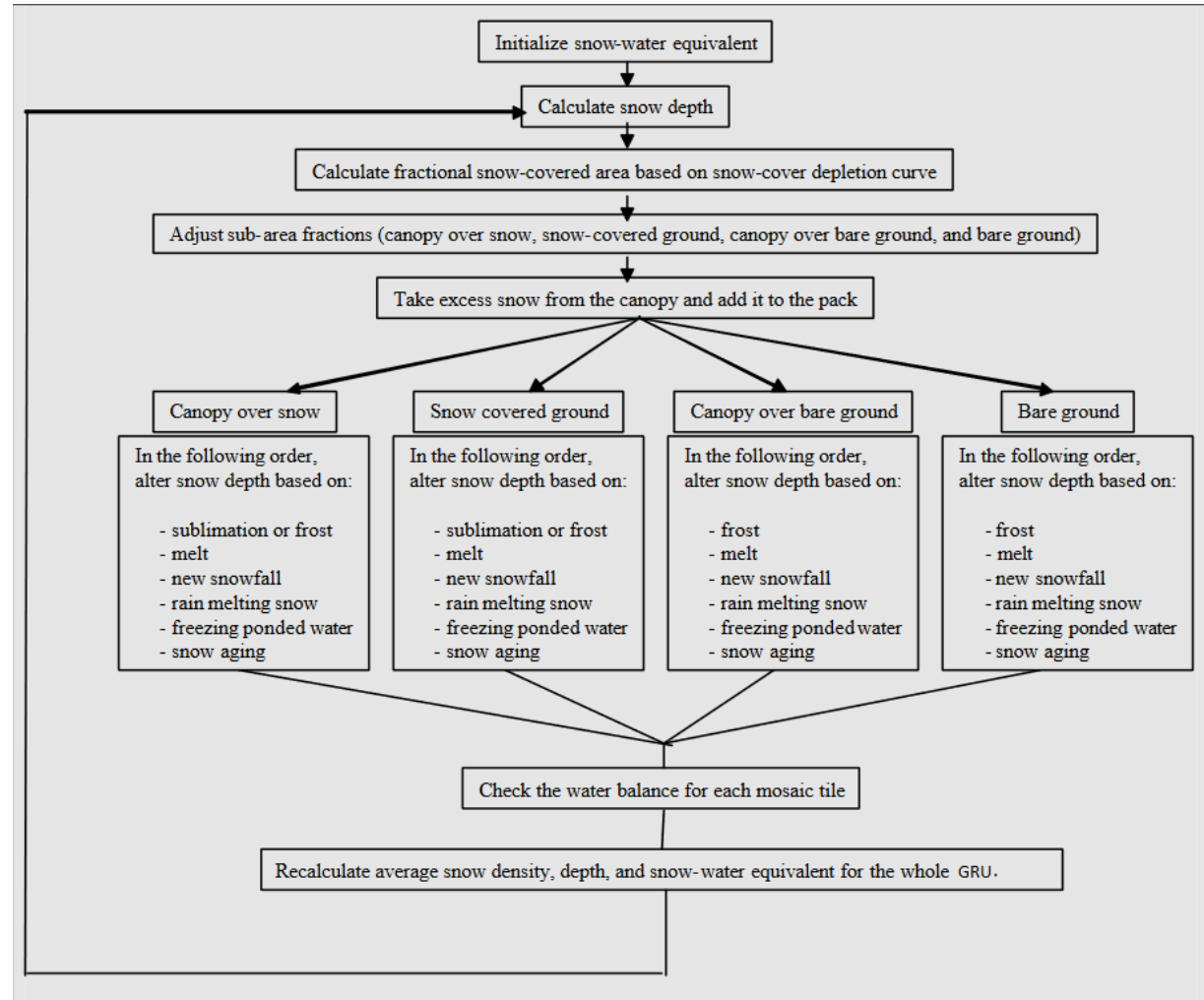
- MESH is the ECCC “Hydrology-Land Surface Scheme”
- Does three things:
 1. Reads atmospheric forcing data on a grid, either from an atmospheric model or from gridded observations
 2. Calculates the vertical water and energy balance at the land surface (LSS)
 3. Calculates lateral hydrologic fluxes on the land surface and routes excess water through a river channel network (hydrology)
- It is therefore a **physically-based, distributed, hydrological model**

MESH as a black box



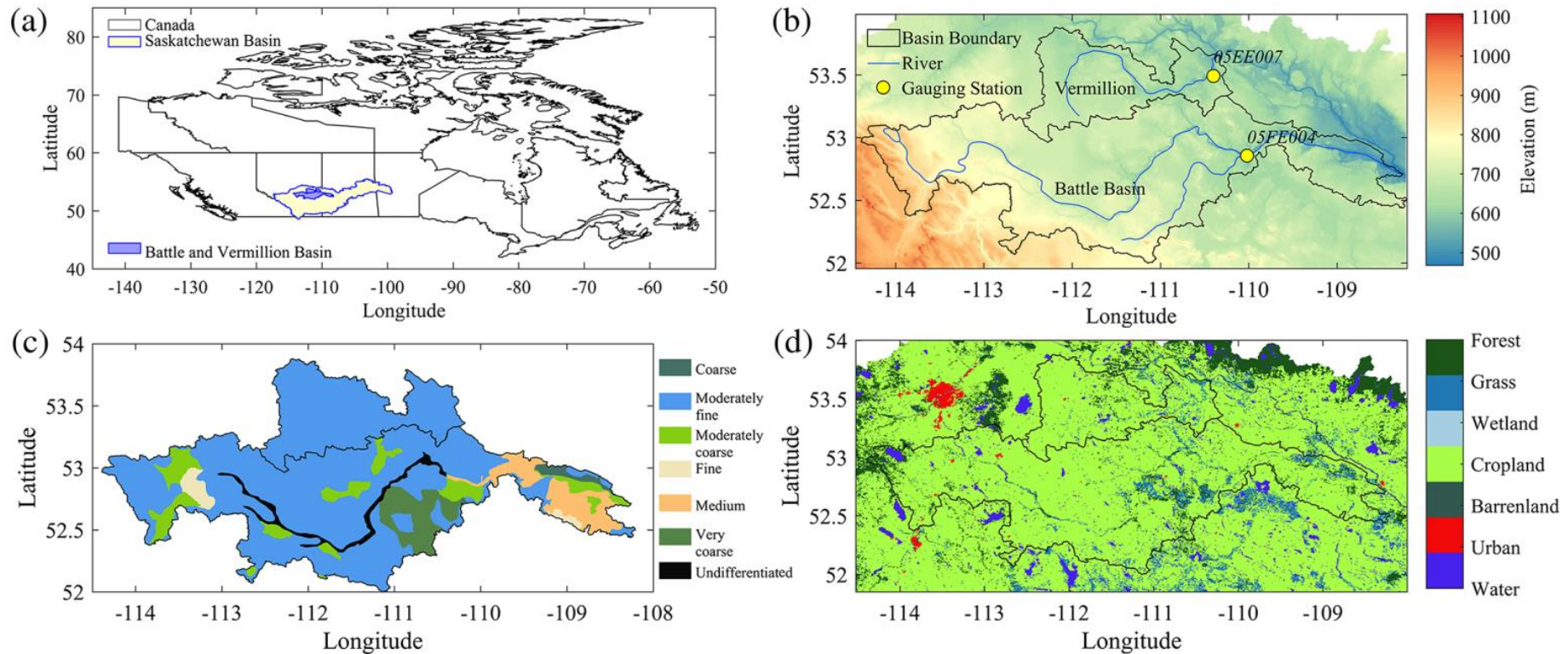
Snow processes

- Snowfall accumulates as a snowpack
- Energy balance based snowmelt
- Other detailed processes, including rain melting of snow, etc.



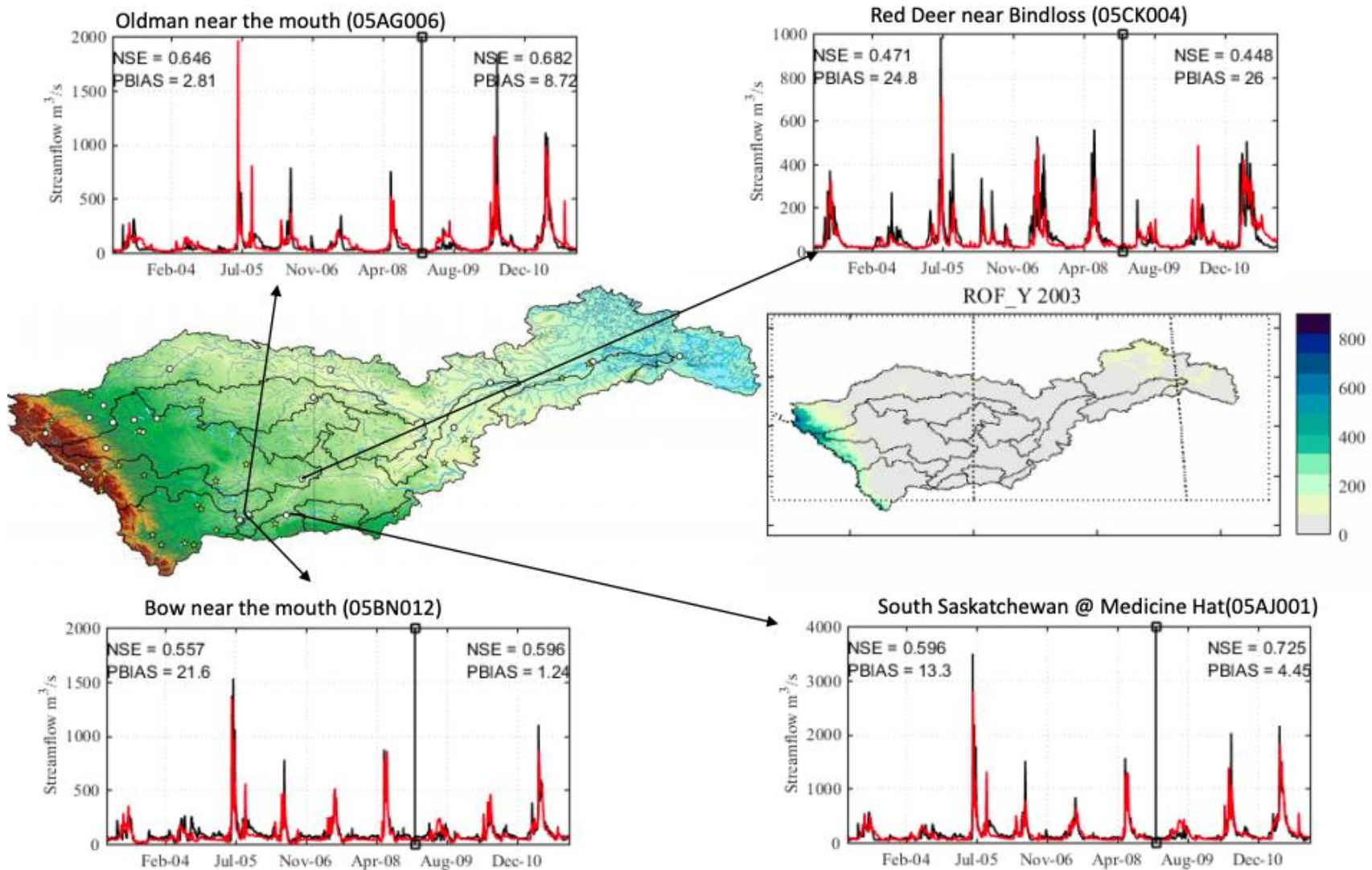
MESH model of the Saskatchewan River Basin

(405,000 km²)



(a) The location map of Saskatchewan River Basin (SaskRB) and its Battle and Vermilion subbasins. (b) Battle and Vermilion subbasins DEM, boundary, and gauging stations. (c) The map of soil parent material texture groups. (d) The map of landcover

Modelling the Saskatchewan River Basin



MESH disadvantages

- Requires relatively more training to be able to set up and run effectively
- Runtimes can be relatively long, depending on how the forecast system is setup, but this is generally not an issue for short simulations of a few days
- Computer power requirements are relatively high, which is a particular problem when calibrating the model
- Because MESH inputs are gridded, environmental values measured at stations have to be gridded before they can be incorporated in the model. The gridded data files can be very large and time consuming to produce
- Because MESH is physically-based, it requires more forcing data variables, such as solar radiation, specific humidity, and wind speed than do conceptual models

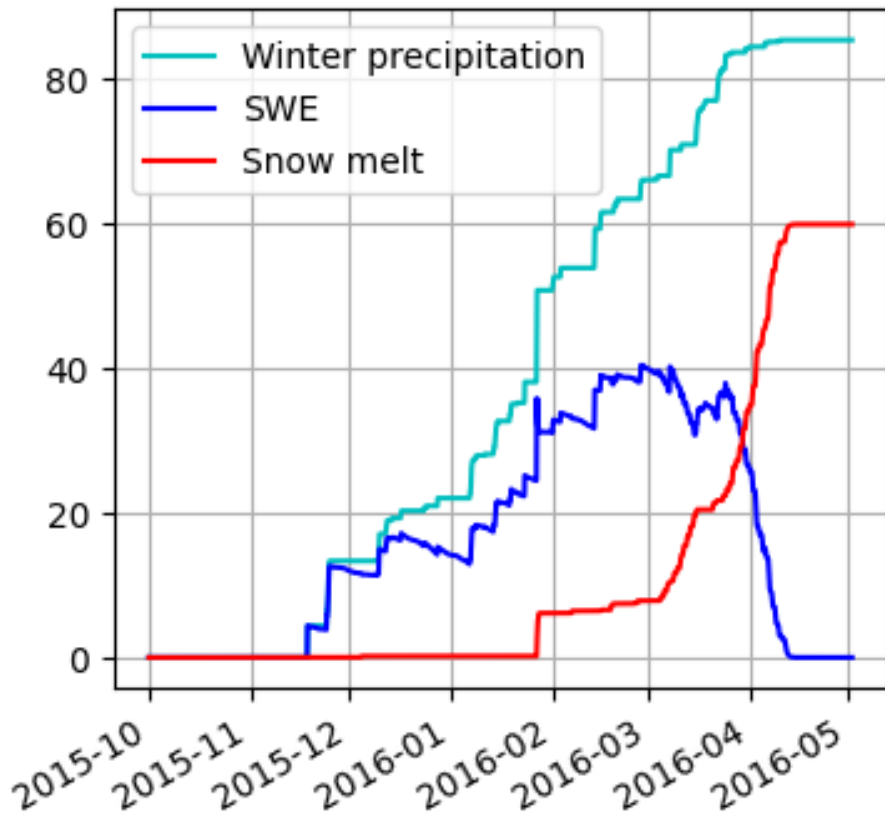
MESH advantages

- Can be integrated with weather and climate models e.g. to simulated climate change impacts
- Works very well with reanalysis data and with short-term weather forecasts, as is being done in the Yukon
- More sophisticated than most other models for application in cold regions
- Operational river basin models will require comparatively little adjustment of snowpacks
- Can be readily modified (by GIWS/ECCC) to address any particular user needs
- Can assimilate gauged streamflows
- Simulates more than just streamflow - for example gridded SWE and soil moisture can be output
- Can be applied in ungauged basins
- Less vulnerable to errors due to the spatial variability of precipitation than are lumped or semi-distributed models

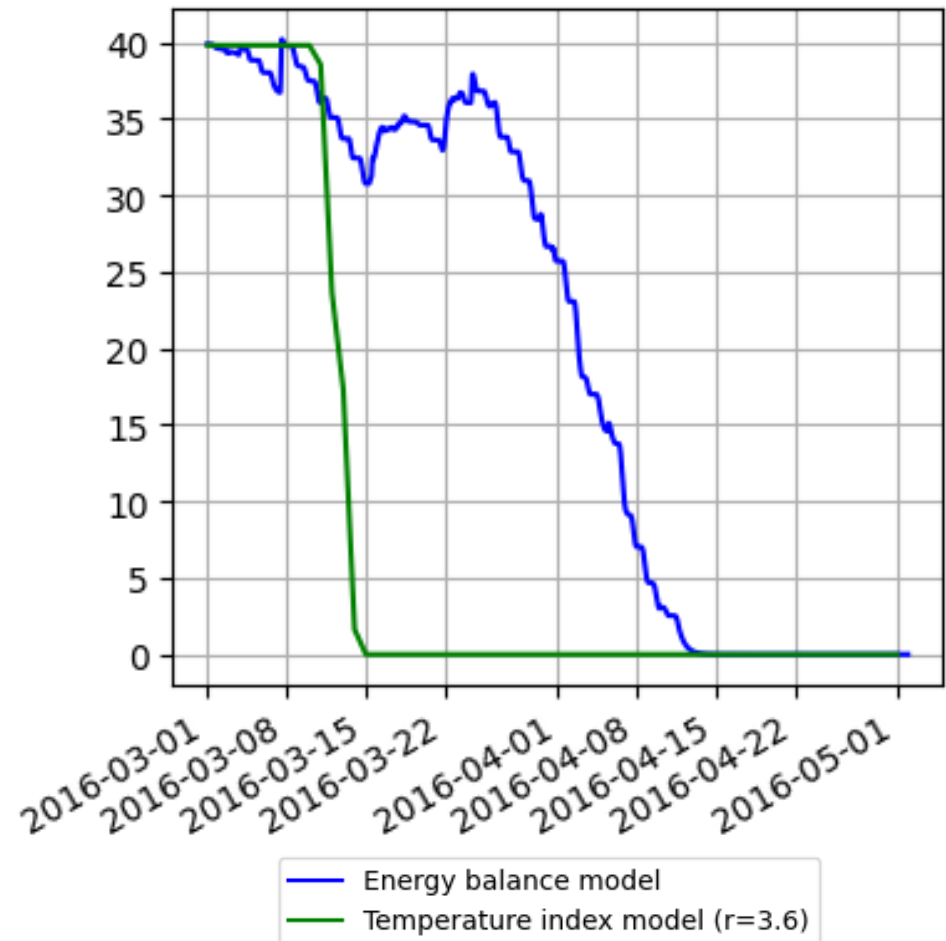
COMPARING MODELS

Energy balance vs temperature index snow models

Energy balance model:

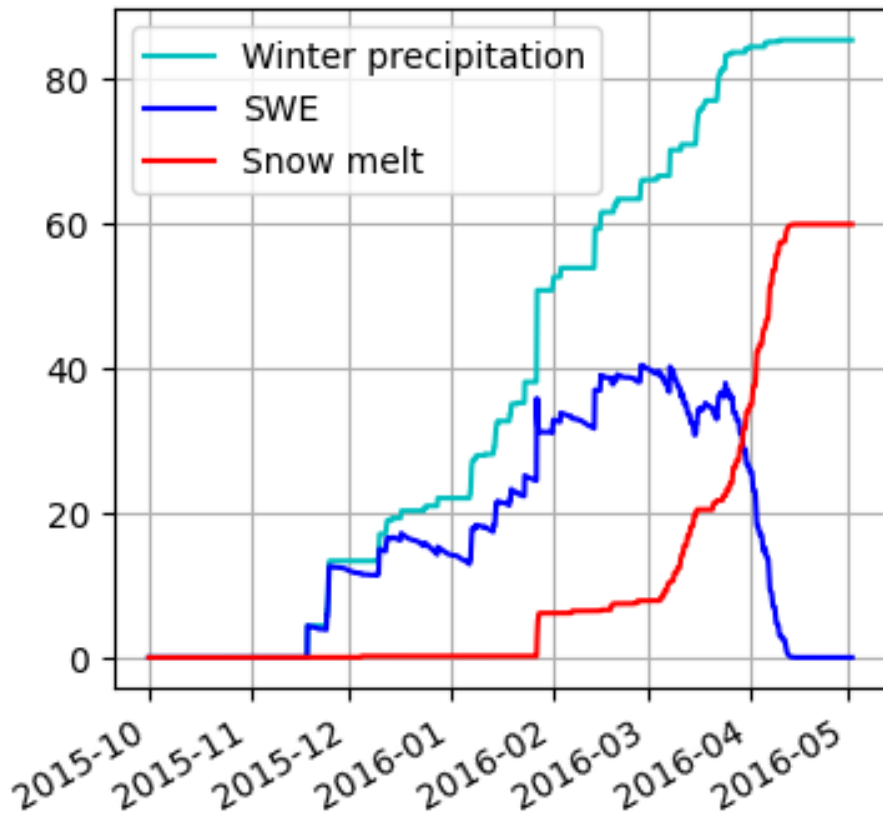


Melt with default parameters:

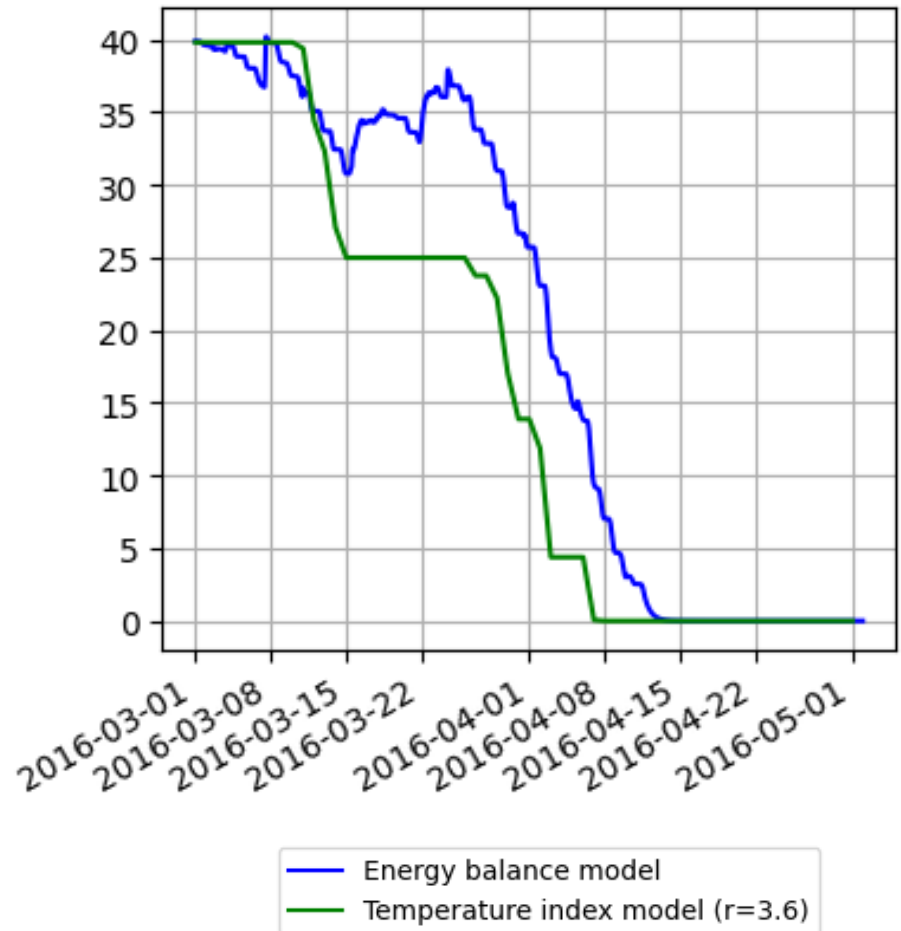


Energy balance vs temperature index snow models

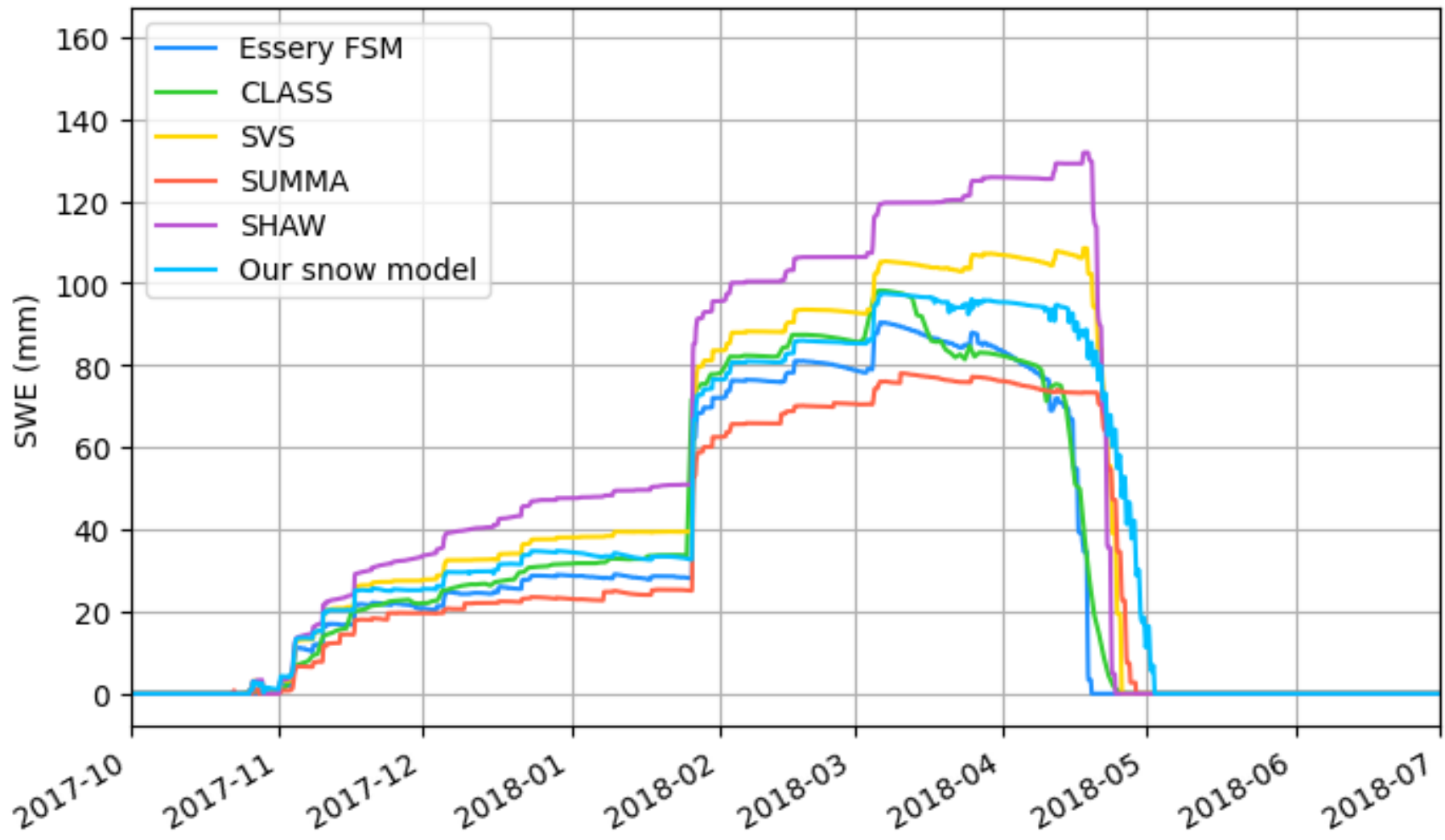
Energy balance model:



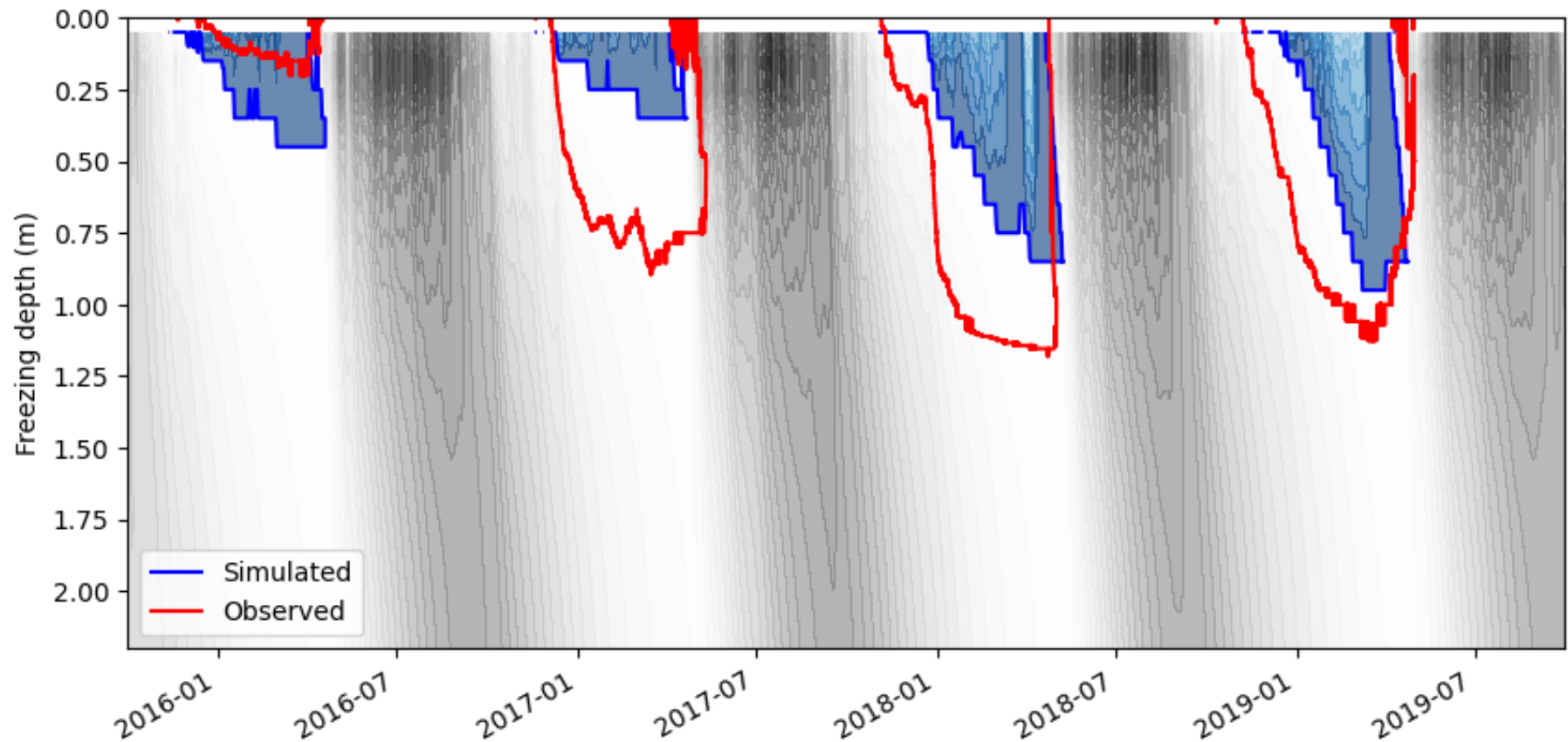
Calibrated temperature index model



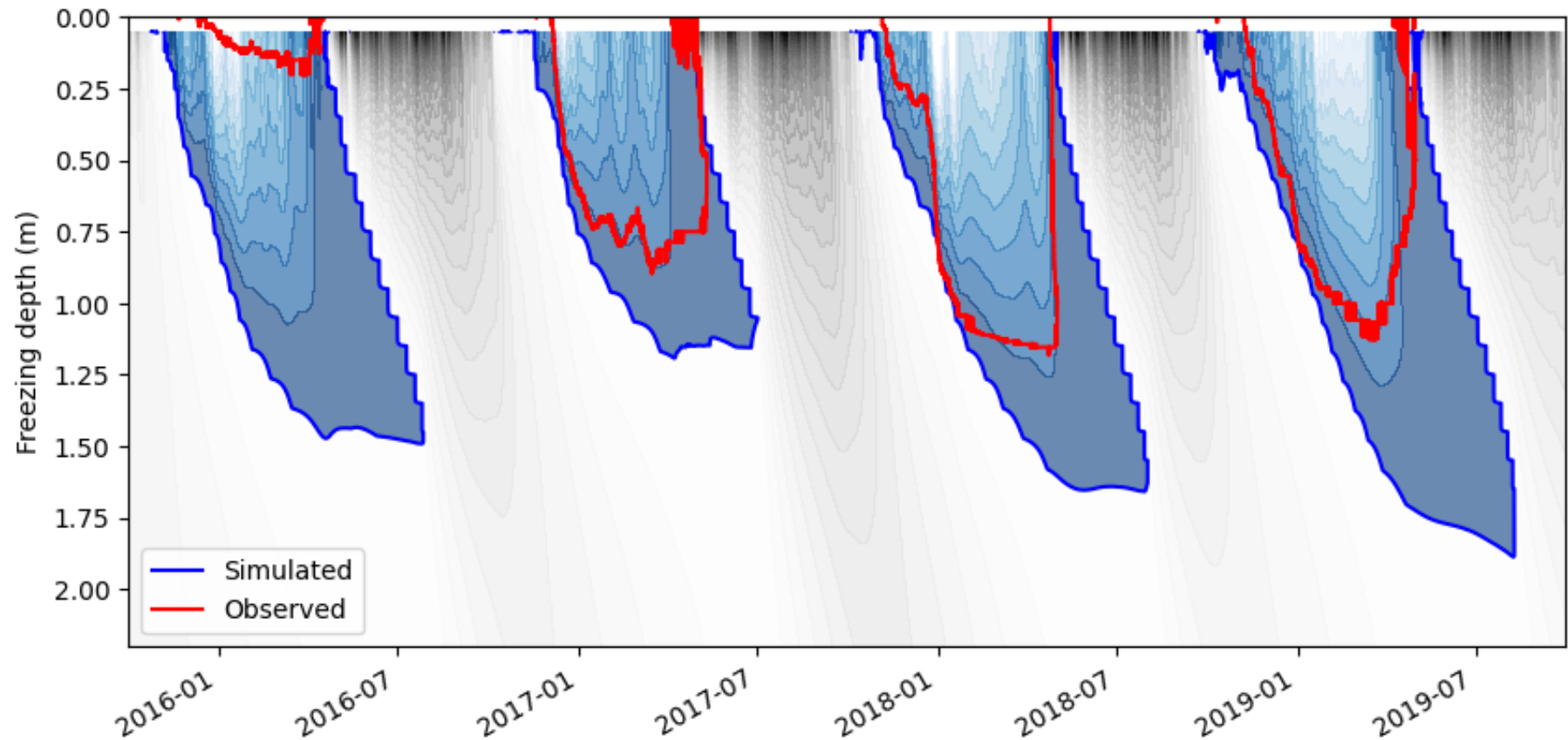
Comparing alternative LSS snow models



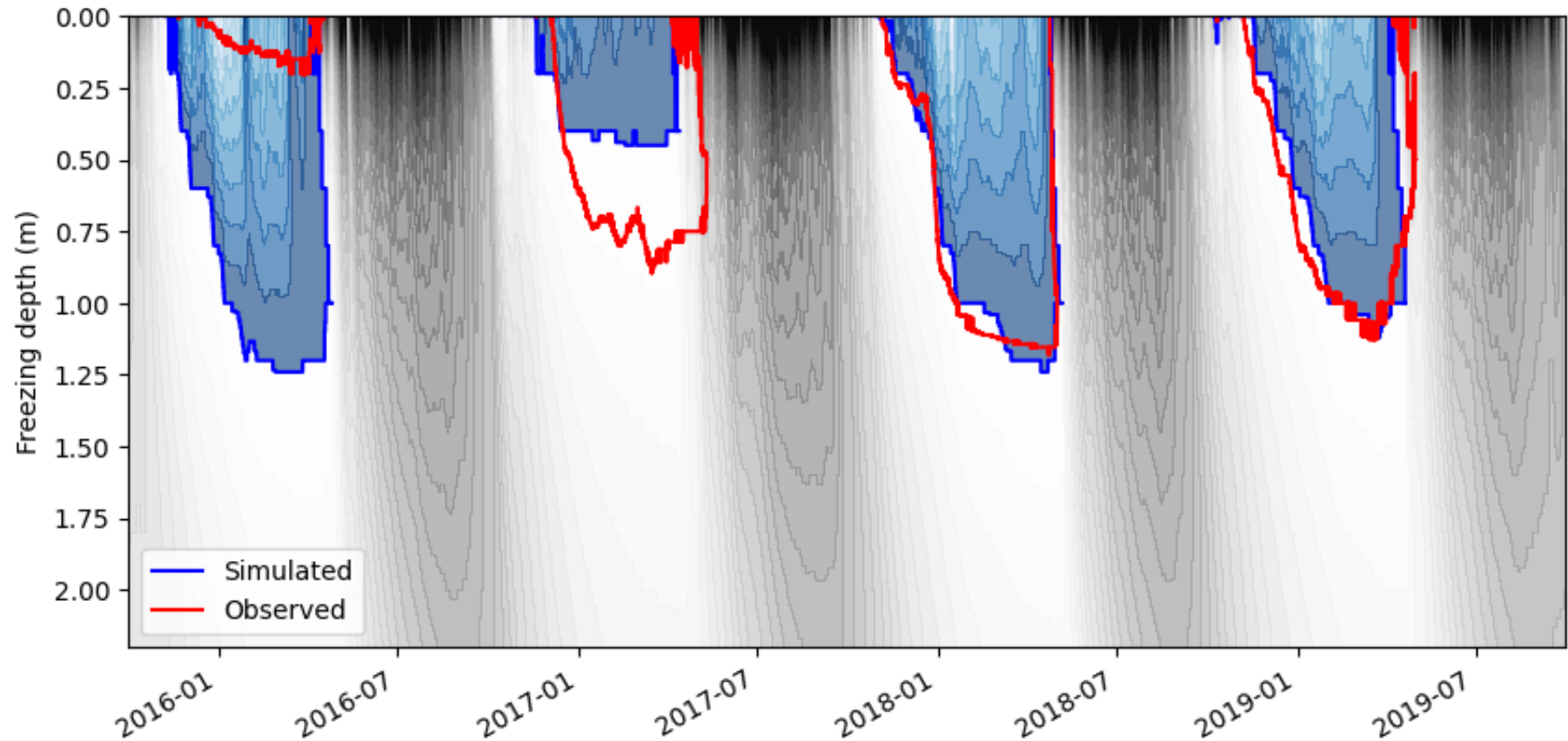
Comparing soil freezing in different LSS models: CLASS



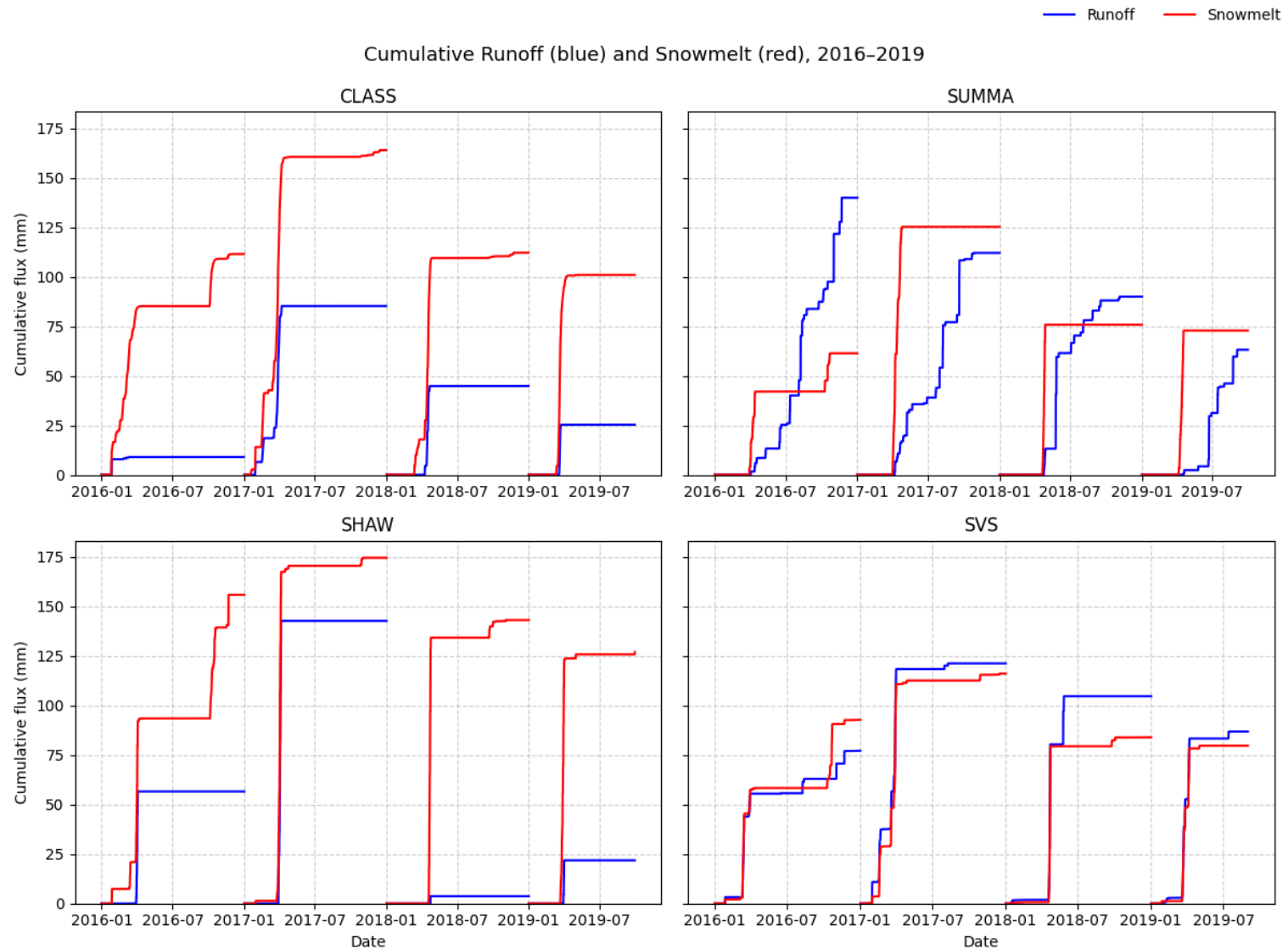
Comparing soil freezing in different LSS models: SUMMA



Comparing soil freezing in different LSS models: SHAW



Comparing snowmelt and runoff from different LSS models



Summary:

- Most hydrological models are not developed to be applied to cold regions where snowmelt dominates
- Cold regions hydrological models include conceptual models and physically based models
- Conceptual models are simpler and often use a temperature index for snowmelt. This is over-simplified and the models are likely not good for climate change simulations
- Physically-based models include land surface models as one example, and there are many different LSS in existence.
- The advantage of physically based model is that they model the energy balance – hence snow melt processes can be well represented.
- The disadvantage is the size and number of parameters in the model
- One critical process that is still poorly modelled is partitioning of snowmelt between runoff and infiltration
- Different physically based models simulate quite different outcomes – meaning we still have work to do to get the processes right